

Mathematical and Computational Techniques for Multilevel Adaptive Methods

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Ergreife die Feder

*Das Mögliche ist ungeheuer. Die Sucht
nach Perfektion
zerstört das meiste. Was bleibt
sind Splitter
an denen sinnlos gefeilt wurde.*

Friedrich Dürrenmatt

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This monograph is an attempt to present the basic concepts of fully adaptive multilevel methods, including their mathematical theory, efficient algorithms, and flexible data structures. All these aspects are important to obtain successful results for practical problems. Additionally, I hope to show with this book that a unified approach combining these aspects leads to many new insights.

Multilevel adaptive methods have evolved rapidly over the last decade, and the development has reached a point where the different aspects of the discipline are finally coming together. This book is meant to be a reflection of this maturing discipline. However, the attempt to present all components of adaptive methods from functional analysis to software engineering within limited space and time has forced me to make compromises. Therefore, I have tried to simplify the material by concentrating on instructive prototype cases instead of representing the full generality of the ideas. The reader may also be warned that, despite my attempts towards unification, the theoretical foundation of multilevel methods in approximation theory requires a scientific language that is different from what is needed to discuss the benefits of object-oriented programming for these methods. Nevertheless, I hope that the selection of topics and the style of representation will be useful to anyone interested in multilevel adaptive methods.

The theory of multilevel methods has been studied systematically for almost three decades, with many papers appearing on the subject, especially in the past few years. It has therefore become difficult for one person to follow all of the different developments. In this monograph, I attempt to present a theoretical foundation of multilevel methods that fits especially well into my view of adaptive methods. In the references, I have included pointers to topics beyond the scope of this book, as they are accessible to me at the time of writing.

Many fewer publications are available on data structures for multilevel methods, and most of them are limited to the implementation of special adaptive strategies. My approach attempts to go beyond this, because I believe that ultimately a more general, abstract treatment of the computing aspects

will be as essential as a sound mathematical foundation.

Many of the ideas presented in this book were developed while I was visiting the Computational Mathematics Group of the University of Colorado at Denver in 1989. This visit, and my support in 1992, when a first version of this monograph was completed, were sponsored by grants of the Deutsche Forschungsgemeinschaft. I am grateful for helpful discussions with all my colleagues at the Technische Universität München and at the University of Colorado. For many comments that have helped me to improve the book, I am indebted to H. Bungartz, W. Dahmen, A. Kunoth, P. Leinen, P. Oswald, H. Yserentant, and C. Zenger. I want to express my special thanks to S. McCormick for his support and encouragement and a very careful review of my manuscript. Finally, I thank T. Gerstner for his help with the illustrations, the staff at SIAM for their friendly and efficient cooperation, and my wife Karin and my son Timo for their great support and patience.

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Ulrich Rüde

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